INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME

CLASS

INDEX NUMBER

BIOLOGY 9648/02

Paper 2 Core Paper

14 September 2015

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**8**
2	**19**
3	**16**
4	**16**
5	**12**
6	**9**
Section B	
	20
Total	**100**

For Examiner's Use

Section A
Answer **all** questions.

1 Fig. 1.1 shows drawings of a cell at various stages in the mitotic cell cycle.

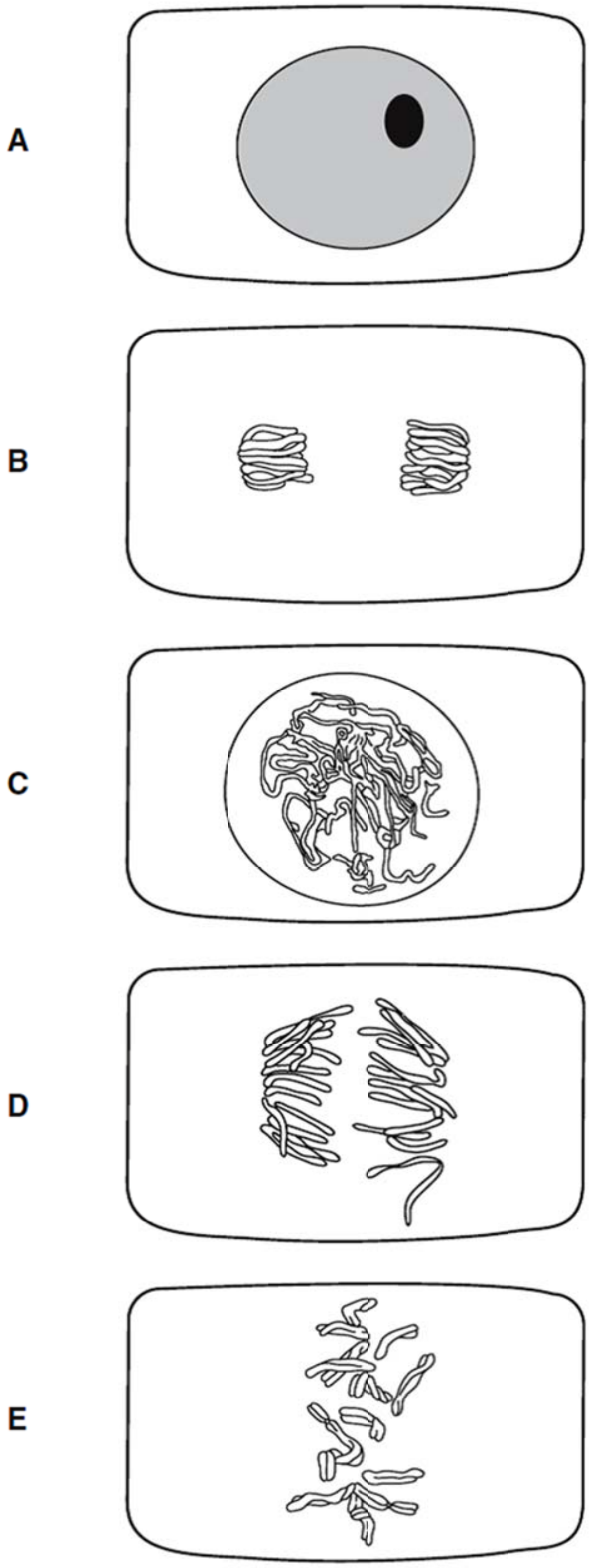

Fig. 1.1

For Examiner's Use

(a) List the letters shown in Fig. 1.1 in the order in which these stages occur during a mitotic cell cycle. The first stage has been entered for you.

A .... [1]

(b) **(i)** Describe events occurring in stage **D**.

....

....

....

.... [2]

(ii) Outline what happens to the DNA in the nucleus during stage **A**.

....

....

....

....

....

.... [3]

(c) State the importance of mitosis in a multicellular organism, such as a flowering plant or a mammal.

.... [1]

(d) The daughter cell from Fig. 1.1 has 8 chromosomes. State the number of chromosomes a cell from the same organism would have at Prophase I of meiosis.

.... [1]

[Total: 8]

For Examiner's Use

2 During immune response, some B lymphocytes change into plasma cells. Plasma cells are fully differentiated B lymphocytes which produces a single type of antibody.

Fig. 2.1 is a drawing made from an electron micrograph of a plasma cell.

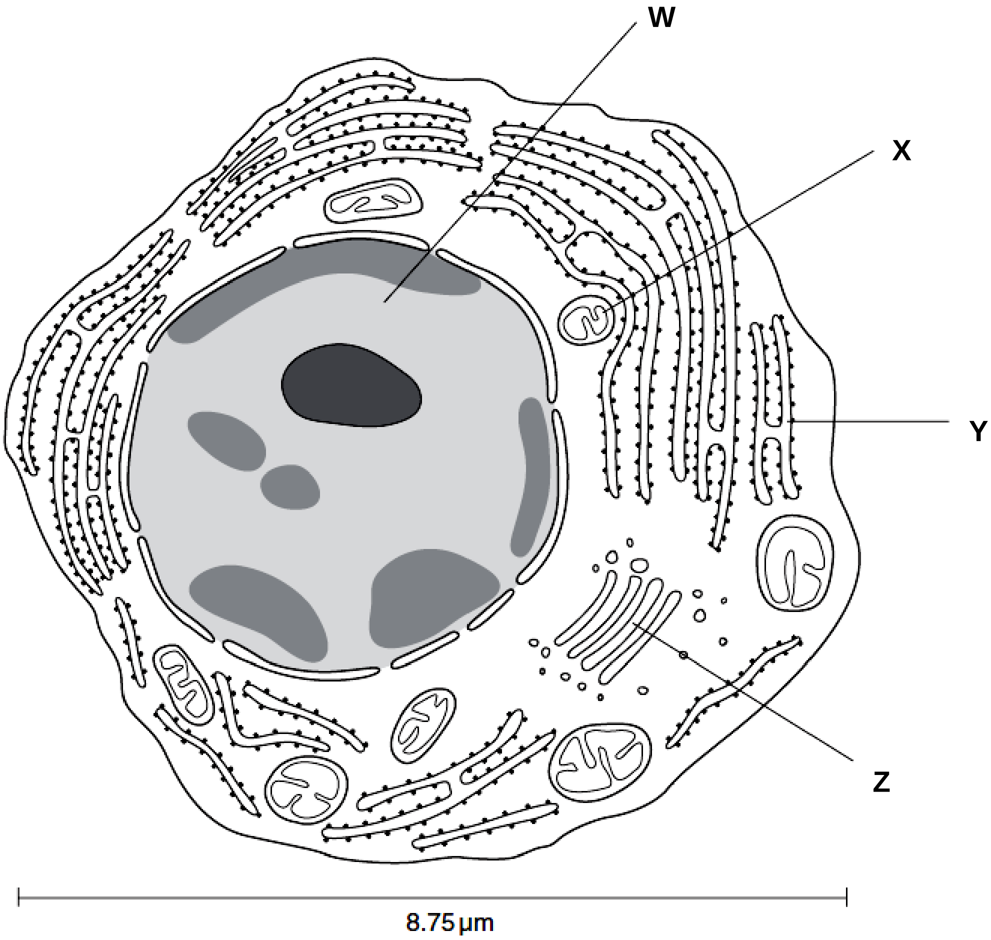

Fig. 2.1

(a) Calculate the magnification of the drawing in Fig. 2.1 to the nearest whole. Show your working.

[1]

(b) Identify structures **W - Z**.

W .... **Y** ....

X .... **Z** .... [2]

(c) Suggest **one** difference in cellular structures between immature B lymphocyte and a plasma cell.

.... [1]

For Examiner's Use

(d) Describe events leading up to the secretion of antibodies after its translation.

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....

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....

.... [4]

Antibodies are proteins produced by B lymphocytes which bind to specific molecules (antigens) on the surface of pathogens like viruses. B cells produce antibodies that are membrane-bound. Each B cell produces a different antibody which recognizes a different antigen.

This antibody, embedded in the cell membrane, protrudes from the B cell acting as a receptor for antigens found on an invading pathogen. During infection, antigens on the pathogen surface bind to the membrane-bound antibody on the B cell thus activating it.

The activated B cell then starts producing antibodies not anchored in the membrane but are secreted into the blood stream where they can aid in fighting infection. Fig. 2.2 shows the different states of B cells.

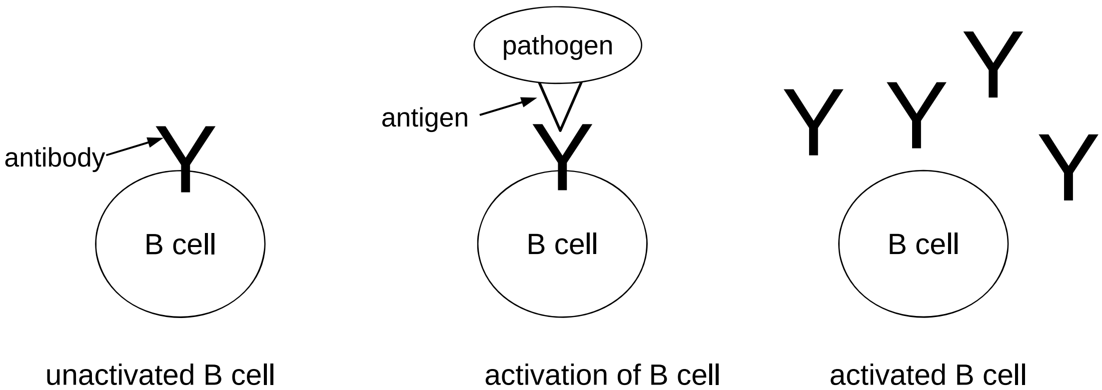

Fig. 2.2

(e) **(i)** Explain how *different antibodies recognize different antigens*.

....

.... [1]

(ii) Suggest how antibodies can *aid in fighting infection*.

....

....

....

.... [2]

The antibody gene contains 8 exons represented in Fig. 2.3.

Legend: E represents exon
I represents intron

E1	I	E2	I	E3	I	E4	I	E5	I	E6	I	E7	I	E8

Fig. 2.3

The functions of the various exons encoded by the antibody transcript are:

- Exon 1 (E1) encodes a signal peptide which helps target the translated protein into the lumen of the rough endoplasmic reticulum.
- Exon 2 (E2) encodes the variable region which is the part of the antibody that recognizes and binds its specific antigen.
- Exons 3 to 6 (E3, E4, E5, E6) encode the constant region of the antibody. This part of the antibody is the same in different antibodies recognizing different pathogens.
- Exons 7 to 8 (E7, E8) encode a transmembrane domain that anchors the antibody in the B cell membrane.

(f) Pre-mRNA transcribed from the antibody gene undergoes post-transcriptional modifications to produce a mature mRNA which is translated into antibody. Different forms of antibodies can be produced from the same gene by alternative splicing by spliceosomes.

With reference to Fig. 2.2 and 2.3, illustrate in the space provided

(i) a pre-mRNA coding for antibody in an **unactivated** B cell;

[2]

(ii) a mature mRNA coding for antibody in an **activated** B cell after post-transcriptional modification.

[2]

For Examiner's Use

(g) **(i)** State **two** advantages of alternative splicing in eukaryotes.

[2]

(ii) Describe how spliceosome produces a mature mRNA.

[2]

[Total: 19]

3 Fig. 3.1 shows the arrangement of genes in the lac operon.

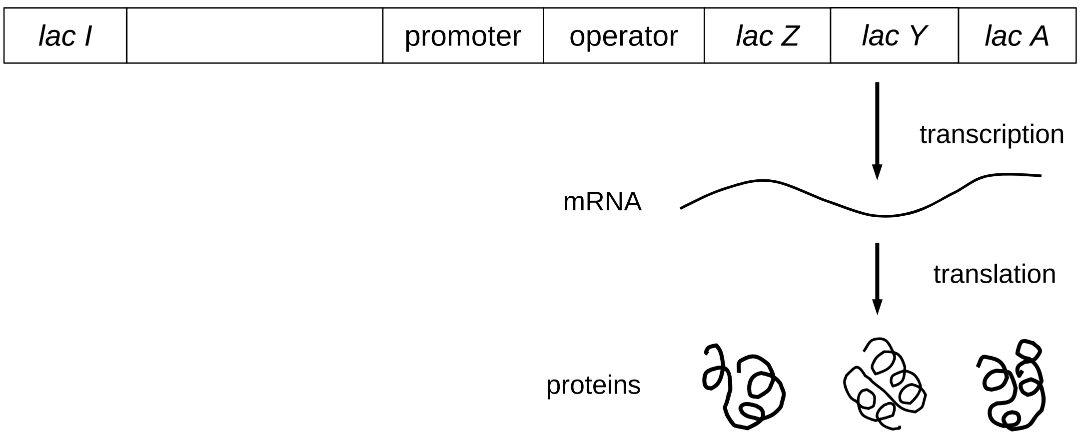

Fig. 3.1

(a) **(i)** Explain what is meant by an *operon*.

[2]

For Examiner's Use

(ii) Name the proteins coded by *lac I, Z, Y* and *A*.

lac I ....

lac Z ....

lac Y ....

lac A .... [4]

(b) Describe what happens in the presence of glucose and lactose in the nutrient medium.

....

....

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.... [4]

(c) Explain how a single mRNA produced from transcription of *lac Z, Y* and *A* can be translated into three separate proteins.

....

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.... [2]

(d) The *lac Z* gene, together with the lac promoter, can be inserted into plasmid vectors used to produce genetically engineered insulin. Explain why the lac promoter is inserted instead of insulin promoter.

....

....

....

.... [2]

For Examiner's Use

(e) Another operon, the arabinose operon, consists of structural genes *ara B*, *ara A* and *ara D* which codes for enzymes that break down arabinose. Explain if arabinose is an inducible or repressible operon.

....

....

....

.... [2]

[Total: 16]

4 Glucose-6-phosphate dehydrogenase (G6PD) deficiency is a sex-linked recessive genetic condition. G6PD is involved in the conversion of glucose-6-phosphate to ribose-5-phosphate, producing reduced nicotinamide adenine dinucleotide phosphate (NADPH) in the reaction. NADPH reacts with harmful molecules called reactive oxygen species, protecting cells from damage and destruction.

The production of NADPH by G6PD is especially important in red blood cells, which are particularly susceptible to damage by reactive oxygen species because they lack other NADPH-producing enzymes. G6PD deficiency causes premature red blood cell lysis leading to haemolysis.

(a) **(i)** Explain what is meant by *sex-linked* disease.

....

.... [1]

(ii) Using the information given, explain why only homozygous recessive individuals experience haemolysis.

....

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....

.... [3]

For Examiner's Use

(b) Fig. 4.1 is a pedigree showing inheritance of G6PD deficiency in a particular family.

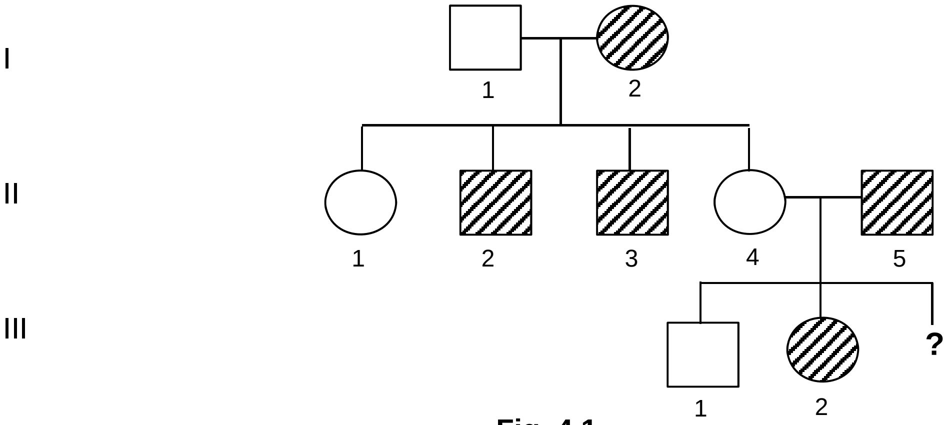

Fig. 4.1

(c) With reference to Fig. 4.1, cite evidence that the inheritance of G6PD is sex-linked recessive.

-- [2]

(d) **(i)** Blood group of several individuals are as follows:

I1 blood group A
I2 blood group B
II4 blood group A
II5 blood group O

Using a genetic diagram, illustrate the probability that **II4** and **II5**'s third child would be a daughter with blood group A, who suffers from G6PD deficiency.

Legend

X^G	normal G6DP	I^A	blood group A
X^g	G6PD deficient	I^B	blood group B
		I^O	blood group O

[5]

(ii) Explain why the probability of the third child having G6PD deficiency is independent of whether the first and second child suffers from the disease.

[2]

(e) **(i)** A chi-squared test was conducted to investigate the occurrence of G6PD deficiency in generation **II**. A χ^2 value of 2.8 was obtained. Using Table 4.1 provided, conclude if the observed ratio deviates significantly from the expected.

Table 4.1

DF	Probability, p				
	0.10	0.05	0.02	0.01	0.01
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82

[2]

(ii) Suggest **one** reason why the conclusion in **(e)(i)** may not be valid.

[1]

[Total: 16]

5 Diabetes mellitus is one of the leading causes of death in Singapore. The International Federation of Diabetes estimates 387 million people worldwide have diabetes, with an increasing trend. There were 533 600 cases of diabetes in Singapore in 2014, with more than 4000 deaths in adults due to the disease.

Diabetes is caused by insufficient or ineffective insulin which functions to lower blood glucose level. Diabetes is a chronic disease with no cure. However, it can be controlled through various means such as a healthy diet, regular exercise, oral medication to balance blood glucose, insulin injections etc.

Insulin, a peptide hormone, is produced by the pancreas when the glucose concentration of the blood is low.

Except in the case of type 2 diabetes, an increase in blood insulin concentration lead to insulin binding to specific sites on the cell surface membrane of target cells. This causes the number of GLUT4 molecules to increase on the cell surface membrane. When blood insulin concentration falls, the GLUT4 molecules are removed from the membrane, making it more impermeable to glucose.

On the onset of type 2 diabetes, the patient's cells become insulin-resistant. Initially the pancreas responds by producing extra insulin. This only partially alleviates the problem of insulin resistance and, in time, overworking of the pancreas leads to their death and subsequently a reduction in insulin production. At this stage the patient may need to receive insulin injections, although this offers only a partial solution.

(a) Suggest why incidence of diabetes shows an increasing trend over the years worldwide.

....

.... [1]

(b) **(i)** State precisely, where in the pancreas, is insulin produced

.... [1]

(ii) State **one** target cell of insulin.

.... [1]

(iii) Identify the *specific sites on the cell surface membrane* that insulin binds to.

.... [1]

(iv) Describe how the number of GLUT4 molecules are increased and removed from the cell surface membrane of target cells.

....

....

....

.... [2]

For Examiner's Use

(v) Explain why removing GLUT4 molecules would make the cell membrane more impermeable to glucose.

-- [2]

Table 5.1 presents the results of an experiment comparing rates of glucose production by a group of people with type 2 diabetes and a control group without the condition, during 23 hours of fasting.

Table 5.1

	rate of glucose production per unit body mass / $\mu mol\ kg^{-1}\ min^{-1}$		significance level
	patients with type 2 diabetes	control group	
total glucose production	11.1 ± 0.6	8.9 ± 0.5	$p < 0.05$
glucose from hydrolysis of glycogen in the liver	1.3 ± 0.2	2.8 ± 0.7	$p < 0.05$
glucose from gluconeogenesis	9.8 ± 0.7	6.1 ± 0.5	$p < 0.01$

(c) **(i)** With reference to Table 5.1, explain the difference in glucose production from glycogen and gluconeogenesis in patients with type 2 diabetes and those without.

-- [3]

(ii) Explain the importance of values of significance level to the experiment.

-- [1]

[Total: 12]

For Examiner's Use

6 One laboratory method for investigating possible evolutionary relationships between mammals uses immunology and is summarised in Fig. 6.1.

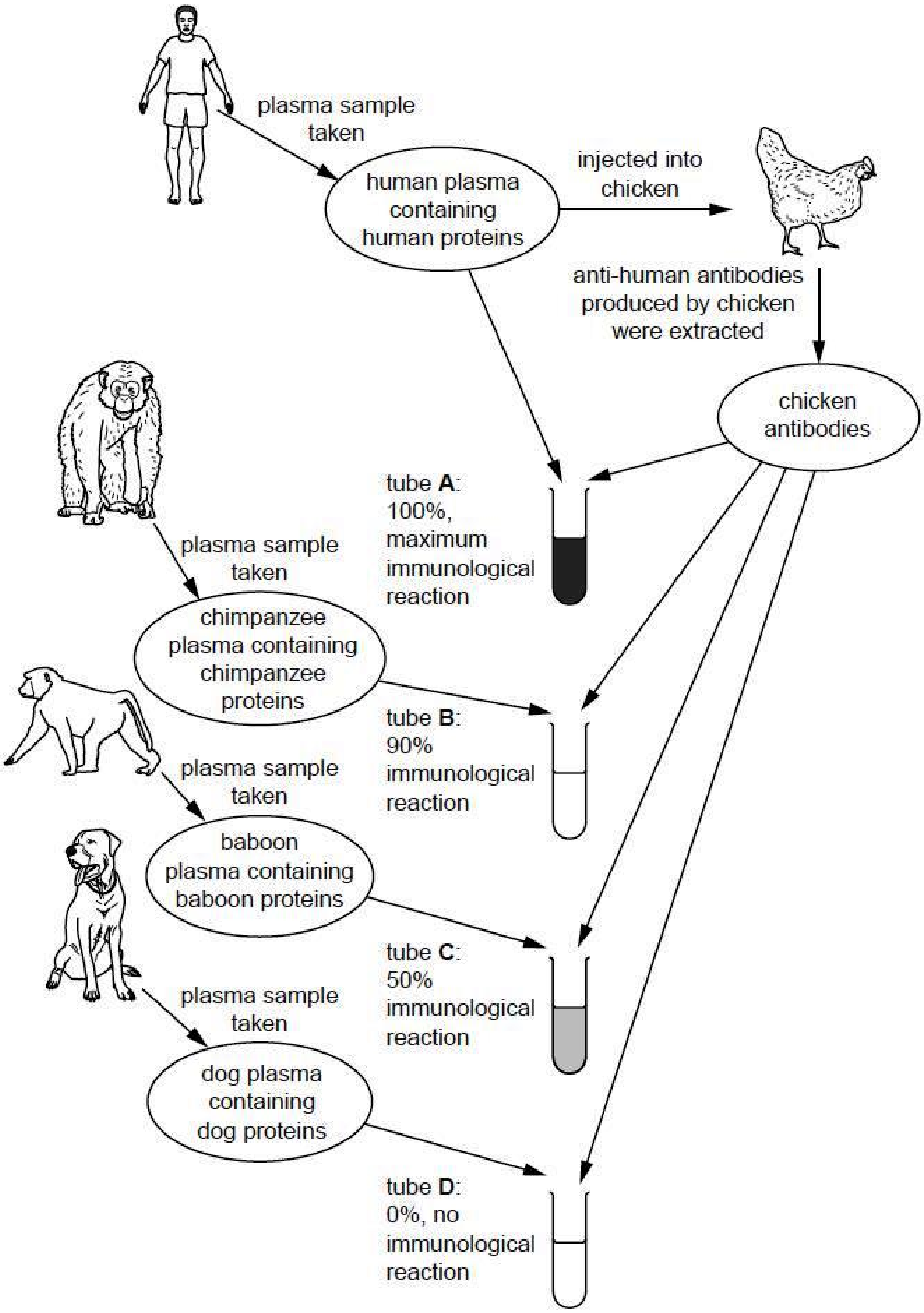

Fig. 6.1

For Examiner's Use

(a) The immune system of birds is similar to that of mammals.

With reference to Fig. 6.1,

(i) explain why injecting human proteins into a chicken causes an immune response.

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....

....

.... [2]

(ii) suggest why investigators described the reaction of the chicken antibodies with the human proteins as 100% (tube **A**).

....

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.... [2]

(b) Describe and explain the conclusions which could be made about the evolutionary relationships between the mammals shown in Fig. 6.1.

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.... [4]

(c) State the highest taxon shared in common by organisms shown in Fig. 6.1.

.... [1]

[Total: 9]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

7 **(a)** Describe the structure of DNA and how it is packed into chromatin. [6]

(b) Describe what happens to the DNA molecule during S phase of interphase. [7]

(c) With a named example, explain how a change in the sequence of the DNA nucleotide may affect the phenotype of the organism. [7]

[Total: 20]

8 **(a)** Carl Linnaeus introduced, in 1735, a system of naming and classifying living organisms just over one hundred years before Darwin published his theory of evolution.

He wrote:

- 'All the species recognized by Scientists came forth from God's hand, and the number of these is now and always will be exactly the same.'
- 'The species are as numerous as the different forms which exist upon this globe – they produce others similar to themselves but in greater numbers.'
- 'Organisms are divided into two kingdoms of nature: viz. the vegetable and animal kingdoms.'
- 'The species and the genus are always the work of nature; the class and the order are the work of nature and art.'

Compare Linnaeus' views of the nature of a species and their classification with those of modern biologists, and account for any differences. [6]

(b) Comment on the statement 'Genetically modified crop plants and animals should be treated as new species'. [6]

(c) Explain how homology supports Darwin's theory of natural selection. [8]

[Total: 20]